

Received 30 April 2025, accepted 19 May 2025, date of publication 23 May 2025, date of current version 2 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3573207

RESEARCH ARTICLE

Machine Learning for Earthquake Emergency Evacuation: Site Selection and Neighborhood Navigation

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ABSTRACT This research is first to introduce a machine learning-based method to enhance the quality and speed of selecting emergency evacuation centers in Tehran, optimizing the use of the city's current capacities. Tehran, with a population of 8.7 million, is located on multiple active faults, where 258 out of its 373 neighborhoods lack sufficient emergency evacuation centers. This inadequacy poses a significant risk during emergencies, potentially leading to failures in the evacuation process. By employing an artificial neural network model informed by data from San Francisco—a city with similar seismic risks, urban structure, building types, and population density—this study achieved a 23% reduction in the average minimum distance from each parcel to the nearest emergency evacuation center compared to the current conditions. Additionally, per capita adequacy at the neighborhood level increased from 27% to 59%. The second goal of the research is to streamline the emergency evacuation process through the development of an application that dynamically navigates users to the nearest evacuation center identified by the model. This application is built using OpenStreetMap and the OpenRouteService routing algorithm. With the successful achievement of both goals, the emergency evacuation process in Tehran is now smarter and more comprehensive. The model here can also be applied to other cities that are similar to the cities studied in terms of site-selecting criteria. For cities with less similarity, the model needs to be slightly trained with the data of the target city to achieve desirable results.

INDEX TERMS Artificial intelligence, earthquake disaster management, emergency evacuation centers, navigator.

I. INTRODUCTION

The seismic vulnerability of Tehran, the capital city of Iran, has adequately been documented in many studies. This city is surrounded by major faults such as the North, Mosha, and Rey fault, all of which can cause earthquakes with a magnitude of 7.0 or more on the Richter scale [1]. Historical records have indicated that the city has experienced major earthquakes approximately once every 150 years; the last significant one occurred in 1830 [2]. This historical information has caused

many seismologists to warn that Tehran is overdue for another significant earthquake.

The consequences of a major earthquake in Tehran are numerous. A study conducted by the Japan International Cooperation Agency (JICA) in collaboration with the Tehran Disaster Mitigation and Management Organization (TDMMO) estimated that an earthquake with a magnitude of 7.2 can lead to thousands of fatalities and casualties, and millions of people homeless [3]. In the face of such predictions, the importance of reliable emergency evacuation centers (EEC) shall be further highlighted. These centers provide safe places for displaced residents on the one hand

The associate editor coordinating the review of this manuscript and approving it for publication was Maurizio Casoni¹.

and serve as post-earthquake aid distribution centers on the other hand. Overall, key considerations for EECs include [4]:

- Capacity and dispersion: The network of EECs should potentially accommodate displaced people. They should be distributed throughout the city to ensure an integrated access to all neighborhoods.
- Structural strength: As these centers act often as important buildings, their structural systems must be robust enough to withstand earthquakes larger than the design earthquakes for conventional buildings.
- Critical services: Every EEC must be equipped with essential needs such as water, food, sanitation facilities, and light medical equipment. Power generators and communication systems are also critical to maintaining operations during infrastructure disruptions.
- Multi-purpose functionality: In times of non-emergency, these centers can serve as community centers, sports, or educational facilities, ensuring ongoing maintenance and familiarity with residents.

The effectiveness of EECs is highly dependent on their location where several factors are considered. For instance, they should be located on stable ground and away from areas that may experience liquefaction and landslides [5], [6]. As well, they should be evenly distributed throughout the city to avoid overcrowding while ensuring that all residents have at least one nearby option [5].

As documented, Tehran, with a population of about 8.7 million people according to the official census of 2015 [7], has about 56×10^6 m² of emergency accommodation space. According to the per capita, emergency accommodation space (regardless of the disaster type as stated in the UNHCR standard) should be at least 4.5 to 5.5 m² (≈ 5.0 m²) per person [8]. Examining the current conditions of EECs of the city shows that 13 neighborhoods out of 373 have no center, and 258 neighborhoods lack centers for their populations; this shows the inadequacy of the current condition of EECs in the city. Also, due to the possibility of not being able to operate some EECs due to, for example, destruction of structures or blocking of roads leading to evacuation centers or secondary hazards, there are serious problems in providing enough EECs. Other factors, such as the arrival of a large number of daily commuters, as well as increasing in the city population, can aggravate the problem of providing the required EECs.

It is understood that providing functional guidance to fully evacuate residents from high-risk areas to EECs, in a way that the utmost safety is met, is an ideal goal. This, however, can be a very challenging task for populated cities such as Tehran. The challenges can even be exacerbated by potentiality of the buildings destruction, collapse, and routes blockages.

In the specific case of Tehran, at least, two reasons can impede the execution of a full evacuation, as follows.

1. Currently, alongside numerous highways and main streets stretched throughout the city from north to south and west to east, there are additionally many corridors that are

exclusively employed for emergency and security vehicles; no private vehicle is yet allowed to use them. As a post-earthquake traffic jam is expected on most roads, a full evacuation will not be an easy task [9].

2. Some studies have indicated a high probability of road blockage due to the demolition of structures and infrastructure, making it difficult to conduct a full evacuation program [9].

Based on these two reasons, it is rational to adopt a neighborhood-based approach for evacuation and relocation, allowing residents to move from high-risk areas to EECs without relying on long traveling. This goal can be achieved by leveraging Artificial Intelligence (AI), a widely used tool in disaster management, through the application of machine learning (ML) algorithms. In the following sections, the relevant studies on EECs, emergency evacuation, and the application of AI in earthquake disaster management are discussed.

II. LITERATURE REVIEW

A. SITE-SELECTING OF EECs

Trivedi and Singh [10] proposed a hybrid multi-criteria decision-making model for shelter site-selecting using Fuzzy Analytic Hierarchy Process (FAHP), and Goal Programming. The research method involved converting multiple objectives into a single function and using the experiences of decision-makers to decide on shelter locations. The results of this model were confirmed in the Nepal earthquake. Soltani et al. [11] provided a systematic literature review of over 270 articles to appreciate the criteria affecting the site selection of shelters for earthquake victims. The results indicated that criteria such as accessibility and proximity to the homes of the earthquake victims are further emphasized in most studies. Then, they divided these criteria into six main categories: size and location, hazard mitigation, rescue facilities, site capability, and environmental and social aspects. Nappi and Souza [12] conducted research with a comprehensive review of the factors considered in the selection of shelter sites. They examined important criteria in the field of displacement and temporary accommodation of affected residents. The research method involved identifying and hierarchical organizing of ten important qualitative and quantitative criteria that are used in the multi-criteria model. The results showed that the model can enhance logistical decisions at both strategic and operational levels. Xu et al. [13] investigated the site selection of earthquake evacuation shelters using a multi-criteria model. The research method included the selection of candidate locations, analysis of spatial coverage, and determination of final locations using GIS. In another study, Unal and Uslu [14] examined parameters such as distance from medical centers, distance from fault lines, population density, and distance from urban open spaces using geographic information systems. Yu and Wen [15] examined the desirability and adequacy of existing emergency safe evacuation points using GIS and the AHP

method. Wang [16] investigated the desirability of emergency shelters in Tianjin, China, using the AHP method and TOPSIS. Overall, nine factors were selected in three aspects effectiveness, safety, and accessibility. Hosseini et al. [5] proposed a model for site-selecting emergency shelters in earthquake-prone urban areas. Using the AHP method and examining different physical, social, economic, and environmental criteria, the proposed model was applied to Karaj City, Iran. The results showed that despite the existence of sufficient spaces, due to physical and social issues, there was a need to determine new places to respond to needs. Celik [17] in a study analyzed the criteria for selecting shelter locations. By examining the opinions of 9 disaster managers and reviewing various sources, 6 main criteria and 25 sub-criteria were identified. The results showed that the proximity, capacity of the distribution center, sufficient logistics force, access to electricity, distance to the settlement, and natural hazards such as landslides and floods are the most important criteria for site-selecting a shelter. Bakshi Lomer et al. [18] presented a risk-based decision support system for selecting optimal post-earthquake emergency shelter locations. The research method included identifying 18 important criteria by analyzing stakeholders, standardizing criteria, determining their weights with a large group decision-making model, and evaluating suitable geographical locations with the ordered weighted average method. The results showed that distance from the fault, population density, access to green space and quality of buildings were the most important factors. Zhang et al. [19] optimized emergency shelter locations in Deyang City, China, using multiple geographic data, two-stage Gaussian method, K-means clustering, and particle swarm optimization algorithm. The results showed that the spatial distribution of emergency shelters in the city is irrational, while adding 10 new shelters could reduce the number of inaccessible areas by 43.31%.

B. EMERGENCY EVACUATION

Bayram et al. [20] presented a model for site-selecting shelters and allocating evacuation routes during disasters, intending to minimize the total evacuation time. The model included nonlinear mixed integer programming and considering the tolerance of longer routes by evacuees. The results indicated that increasing the number of shelters and tolerating longer routes can reduce evacuation time and enhance the efficiency of the evacuation program. The efficiency was measured by total evacuation time, the percentage of evacuees within the specified time, and the maximum delay. Chen et al. [21] proposed a time-based location model for earthquake shelters to address the different needs of survivors at different evacuation stages. The research method involved optimizing location selection using GIS and examining the impact of financial constraints on the planning and construction of shelters. The results revealed that the quality of earthquake response is influenced not only by the site-selection method, whether based on discrete or continuous multi-criteria decision-making, but also by financial constraints.

Achieving a balance between the available budget and the efficiency of evacuation—measured by indicators such as time, distance, and evacuation cost—is crucial. Chen and Zhan [22] proposed an agent-based model to simulate evacuation scenarios, in which, factors such as road capacity, traffic flow, and individual decision-making were considered. Their findings emphasized the importance of staged evacuations to avoid traffic gridlock in high-density areas. This is worth noting that in the simultaneous strategy, residents are notified to evacuate at the same time, while in the staged evacuation, residents are divided into separate groups where they are organized to evacuate at intervals. Cova and Church [23] pioneered the use of GIS to identify optimal evacuation routes and potential bottlenecks in urban areas. Their study provided a method to systematically identify neighborhoods that may experience transportation problems during evacuation. Cepolina [24] presented a multi-stage optimal modeling for evacuation. The model was used to record the density along the route. The model assumed that only a set of pre-defined evacuation routes could be used by evacuees. Johansson and Helbing [25] used the Genetic Algorithm (GA) to optimize the location of obstacles to achieve efficient evacuation. Park et al. [26] proposed an optimal routing algorithm for evacuation systems. The network information, in the form of movable spaces in the buildings, was used to determine the optimal route. To design an emergency evacuation program, Naghdi et al. [27] first located suitable places for population settlement using GIS and image processing technologies, and then allocated specific building blocks for each center. Using ML algorithms and GIS, Rahman et al. [28] optimized the distribution of population evacuation centers in Bangladesh to reduce human casualties and increase the speed in the response phase of disaster management. Yabe et al. [29] in a review article, confirmed that mobile phone location data have a great potential to support disaster response, including the evacuation process and population displacement. Idoudi et al. [30] introduced a method for dynamic population evacuation. Initially, evacuation routes were planned, and subsequently, updated traffic information was transmitted to vehicles using vehicle communication networks, enabling route optimization based on changing conditions. The results demonstrated that this approach reduces evacuation time and enhances network efficiency during emergencies. Yousefi et al. [31] suggested a framework to facilitate the rescue and evacuation of victims trapped under the rubbles. Integrating GIS and smartwatches, this framework operated in two phases of before and after an earthquake. First, road blockages were simulated and then, the priority of evacuating the victims was determined based on the health and location data obtained through smartwatches.

C. APPLICATION OF AI IN EARTHQUAKE DISASTER MANAGEMENT

Bose et al. [32] proposed an earthquake early warning system that used deep neural networks to quickly estimate earthquake

intensity and location. Their method combined the advantages of regional and local warning methods. Li et al. [33] introduced a model combining traditional seismological methods with AI techniques, leading to faster and more accurate early warning systems. Ochoa and Comes [34] developed an ML-based model to estimate post-earthquake damage and casualties using satellite images. The estimation was based on a pre-existing database on which the model was trained for determining and establishing shelters. Although the models' accuracy was about 62%, it was a significant step in the field of post-disaster image processing. Sun et al. [35] reviewed the applications of AI in disaster management in all phases of prevention, preparedness, response, and recovery. The research method included the examination of 26 AI-based methods in 17 applied fields of disaster management. The results indicated that most AI-based methods mainly focus on the disaster response phase, and that, the can significantly enhance the speed and volume of data analysis. The research also pointed out the challenges and opportunities of using AI in disaster management and encouraged the improvement of the accuracy and speed of analysis and the development of more cost-effective AI tools for decision-making. Alruqi and Aksoy [36] investigated the use of AI in managing disasters and reducing their effects. The results indicated that the application of AI in disaster management is still in its beginning due to challenges such as the need for high-quality data, high costs, and insufficient awareness. There is a need for greater awareness and increased international cooperation to overcome these obstacles.

Reviewing previous research on site selection for EECs indicates that site selection typically involves gathering expert opinions through questionnaire methods and analyzing the data using various analytical hierarchy processes. In the field of emergency evacuation, the reviewed studies primarily rely on mathematical modeling, agent-based simulations, and optimization algorithms. The main focus of these studies has been on reducing evacuation time, enhancing network efficiency, and improving decision-making during emergencies. Some research has also utilized mobile data and machine learning to update routes and allocate resources. Studies on AI applications in earthquake disaster management have predominantly employed machine learning for data analysis and pattern recognition in earthquake early warning, damage assessment, and post-disaster management. The results emphasize improving accuracy and speed in prediction, damage assessment, and decision-making during the disaster response phase. Generally, in studies, the use of machine learning has not been focused on the domain of emergency evacuation and EECs, particularly for Tehran.

D. THE GOALS AND INNOVATIONS OF THE CURRENT RESEARCH

Given that AI-based tools can effectively be employed for managing EECs site selecting, here, these tools are used for

TABLE 1. Effective criteria for selecting EECs [11], [18], [40], [41], [42].

N o.	Criterion	Required data
1	Land slope	Topography map
2	Land use	Land use map
3	Land area	Land use map
4	Minimum distance to emergency water supplies	Location map of emergency water supplies
5	Minimum distance to medical centers	Location map of medical centers
6	Minimum distance to fire stations	Location map of fire stations
7	Minimum distance to public security centers	Location map of public security centers
8	Population density	Map of population distribution

Tehran City as a case study. For doing this, the experiences of other earthquake-prone cities are used.

By reviewing previous research as given in Section II-A, the use of multi-criteria decision-making methods such as the AHP, which is the result of data collection in the form of a questionnaire, is evident. In these methods, first, a list of criteria is gathered, and then, after preparing a questionnaire and collecting experts' opinions, the results are analyzed. Next, by using statistical methods and weighting the information layers, priorities are given. However, such an attitude can lead to errors due to, for example, the limitation of the statistical population of the respondents, the spatial limitation of the questionnaire results, the non-dynamic nature of questionnaires, and the slow speed of updating the results [37], [38].

Since natural disasters are examples of big data, for effective management, using AI-based models can be very useful. Currently, AI-based methods are used for improving different risk management processes such as exposure, vulnerability, hazards, and risks, as well as prioritizing resources during the response and recovery phases [39]. Additionally, they can be employed in satellite image processing to identify suitable areas for population evacuation, or to optimize conventional site-selection methods.

With understanding of the inadequacy of the current EECs in Tehran, the first goal here is to use the current capacities of Tehran more appropriately, and to eliminate the deficiencies by improving the site-selecting EECs with AI-based tools. The method used here, in addition to solving the errors of questionnaire-based methods, can recognize hidden patterns among criteria, and the possibility to use the successful experiences of other cities.

The second goal is to develop a tool that can guide residents from high-risk areas to Emergency Evacuation Centers

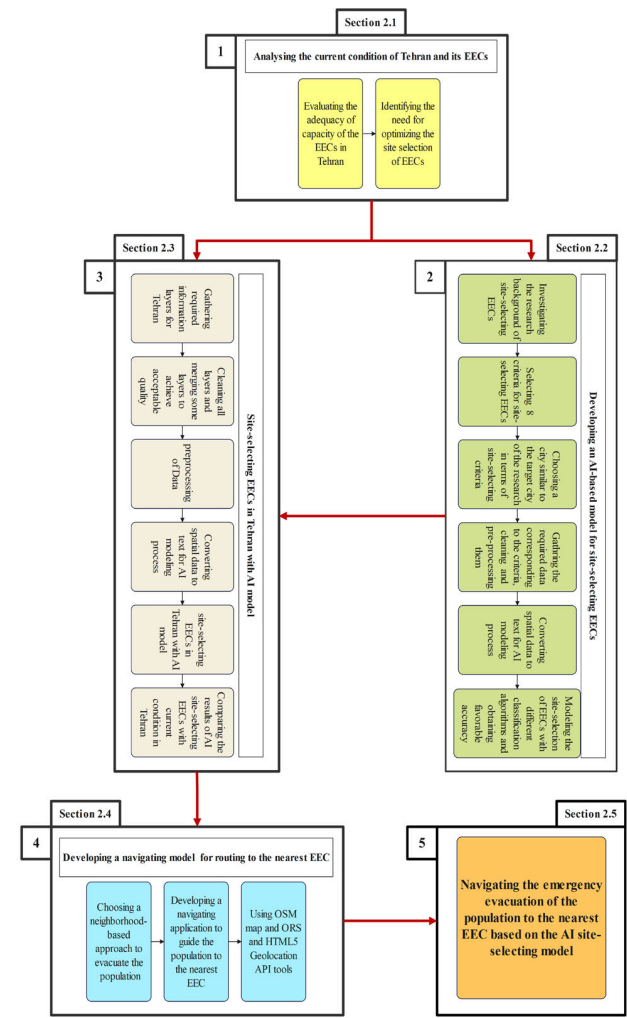


FIGURE 1. Flowchart of research steps.

(EECs). In the context of emergency population evacuation, the direct reliance on the location of EECs and their access routes means that the navigation and evacuation approach using vehicles must incorporate accurate site selection for EECs and an assessment of the vulnerability levels of access routes. However, as noted earlier, using private vehicles during disasters, particularly for long distances, is often impractical. To facilitate evacuation operations for EECs, a neighborhood-based approach serves as an alternative. To integrate the site selection of EECs with the emergency evacuation process and make it complete, an application has been developed to guide users from their current location to the nearest EEC.

To propose a suitable location for EECs, various criteria are involved. Table 1 gives a list of these criteria which have been discussed in many studies.

In Section III, the development of the research method, and in Section IV, the presentation and analysis of the results are discussed. In Section V, the summary, conclusions, and limitations of the research, as well as suggestions for future work are presented.

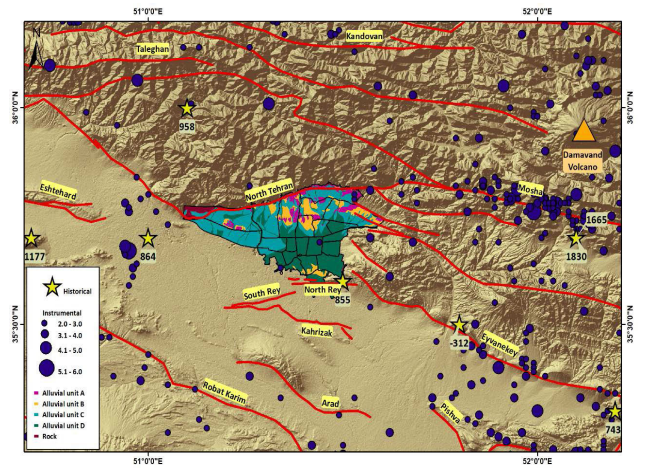


FIGURE 2. Active faults, historical and instrumental seismicity, and alluvial deposits in and around Tehran [45].

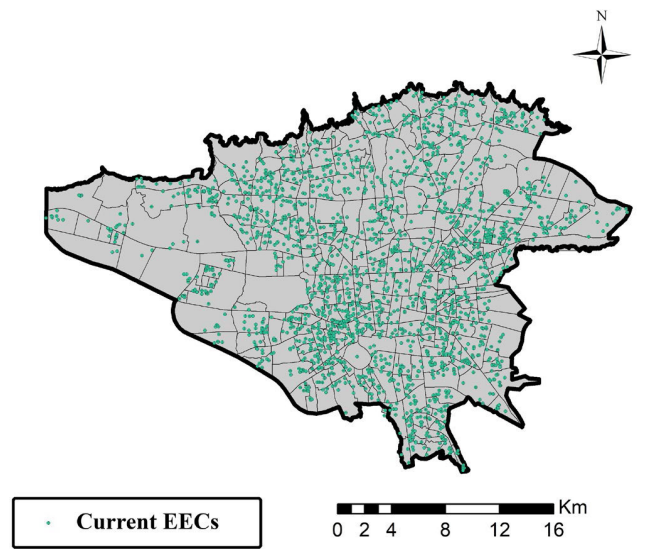


FIGURE 3. The current EECs in Tehran [46].

III. RESEARCH METHOD

Figure 1 shows the steps planned to perform the study here to enhance site-selecting EECs in Tehran and to propose an evacuation process by developing a navigator application. Each step is explained in the following sections.

A. ANALYSIS OF THE CASE STUDY

Tehran has an area of about 730 km² [43]. The population of the city has increased rapidly in recent decades. The status of faults in the city and its surroundings are shown in Figure 2. It is estimated that the peak ground acceleration in these faults can be as large as 1.0g [44].

The current locations of Tehran's EECs are shown in Figure 3. Figures 4 and 5 show examples of current EECs with different land uses. Figure 6 shows some examples of local disaster management centers in the city.

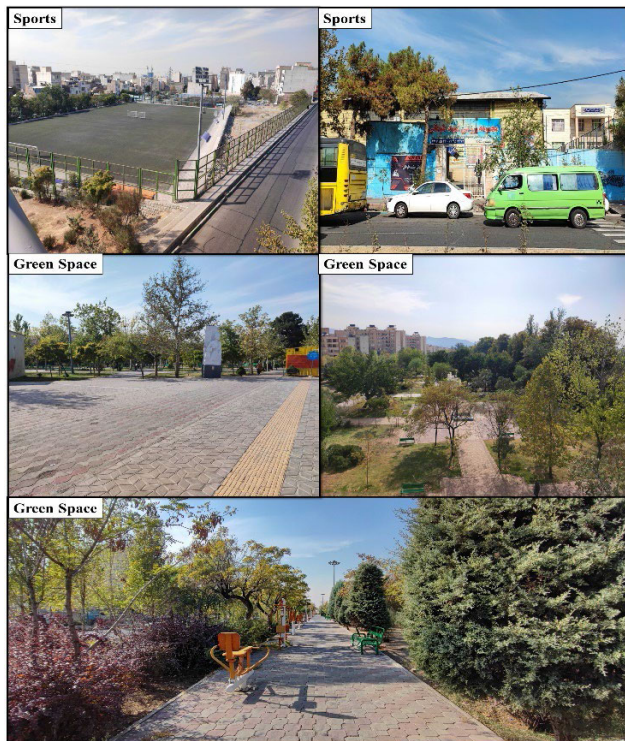


FIGURE 4. Examples of current EECs in Tehran with sports and green space land uses.



FIGURE 5. Examples of current EECs in Tehran with educational land uses.

B. DEVELOPING AN AI-BASED MODEL FOR SITE-SELECTING EECs

Given that the goal of this research is to site-select Emergency Evacuation Centers (EECs) in Tehran, without taking the



FIGURE 6. Examples of local disaster management centers in Tehran.

current EECs into account (Figure 3), it is necessary to choose a city as the base for model training. This city should, first, have successful experience in earthquake disaster management and, second, be similar to Tehran in terms of the eight criteria considered for site selection. If two cities share similarities in site-selection criteria, which serve as the features of the machine learning model, the Cross-Domain Prediction method can be applied to estimate the target values for the target city, Tehran. For this purpose, San Francisco has been selected as the training model's base city. The similarities between these two cities with respect to site-selection criteria are discussed below.

To develop the AI-based models, various algorithms such as Gaussian Processes Classifier, K-nearest Neighbors, Support Vector Machines, and Artificial Neural Network (ANN) can be tested to find which one finally is the most proper. Here, performing detailed analyses on the mentioned algorithms, it was concluded that the ANN is the most proper one which is consequently used for the study planned. To employ the ANN model, the following steps are taken.

- 1) Identifying the effective criteria in site-selecting by examining the background of the research and re-categorizing them according to the research needs; see Table 1.
- 2) Preparing a map of EECs for a sample city with extensive experience in earthquake disaster management.
- 3) Gathering information layers corresponding to each criterion, modifying and extracting data.
- 4) Performing required pre-processing for modeling.

- 5) Developing an ANN model to classify and identify EECs with an F1 score of more than 95%.

Here, to address the second step, the San Francisco city data for site-selecting EECs is employed. Both cities are located in seismically active regions and the maximum magnitude of potential earthquakes can exceed 7.0. Tehran is located in the Alpine-Himalayan seismic belt and faces a significant earthquake risk. San Francisco, on the other hand, is located on the Pacific Ring of Fire and is intersected by active faults, including the San Andreas Fault. The probability of an earthquake with a magnitude of 6.7 or more in the next 30 years in this region is 72% [47], [48], [49]. Great historical earthquakes such as the 1906 earthquake with a magnitude of 7.9 and the Loma Prieta earthquake of 1989 with a magnitude of 6.9 are examples of destructive seismic activity in this city [50], [51].

In terms of urban structures, Tehran and San Francisco have also some similarities. Both cities face many challenges due to rapid population growth and high urban density. The urban modernization of Tehran started in the early 1900s and the renewal of San Francisco was done after the devastating earthquake and pot-earthquake fire of 1906. This historical temporal proximity has resulted in the old fabrics of both cities being similar in age. Both cities also have some dense urban areas, which were formed due to large population and spatial constraints. Additionally, in terms of the maximum and minimum land slope, these cities are very similar to each other, which is effective in infrastructure planning and site-selecting EECs. Both cities also have high population density. Tehran is a city with a population density of more than 10,000 people per km² due to the rapid growth of the city's population [7]. San Francisco also has a significant population density due to its limited land area with a density of more than 6,000 people per km² [52].

The pseudo-code of the site-selecting model to increase the transparency of the research process is presented here:

1. Load and Prepare Data:
 - Read Excel file.
 - Split features (X) and labels (y_raw).
 - Standardize the features.
 - Convert labels to one-hot encoding.
2. Split Data:
 - Split data into training and test sets.
 - Preserve original labels for evaluation.
3. Define Model Builder:
 - Function to construct a neural network:
 - Input layer.
 - Three hidden layers with tunable units.
 - Output layer (Softmax, size 2).
 - Tunable learning rate.
 - Compile with Adam optimizer and proper loss.
4. Set Up Hyperparameter Tuner:
 - Initialize RandomSearch tuner:
 - Use the model builder.

Optimize for validation accuracy.
Set number of trials and storage options.

5. Run Tuning:
 - Train multiple models with different hyperparameters.
 - Early stopping on validation loss.
6. Select and Train Best Model:
 - Retrieve best hyperparameters.
 - Build and retrain model on the training data.
7. Evaluate Final Model:
 - Predict on test set.
 - Compute F1-score, and confusion matrix.
8. Save Final Model:
 - Write trained model to file.

C. SITE-SELECTING EECs

In this section, the developed model is utilized to enhance site-selecting EECs in Tehran. The results are then compared with the current condition of EECs in the city. The information layers of Tehran city that are used are as follows [46]:

- 1) The land use of parcels
- 2) The elevation points map the city
- 3) The location map of public security centers
- 4) The location map of medical centers
- 5) The location map of fire stations
- 6) The location map of emergency water supplies
- 7) The population distribution map
- 8) The map of neighborhoods
- 9) The map of districts
- 10) The map of the current location of EECs in the city

The following corrections and pre-processing are required after data collection to prepare the model input:

- Unifying maps coordinate system to Universal Transverse Mercator.
- Preparing land use maps by merging existing maps.
- Labeling the parcels of the current EECs on the land use map.
- Calculating the area of each parcel.
- Centering and determining the coordinates of the center of each parcel for future textual output of information layers.
- Calculating the minimum distance (Manhattan) of each parcel from medical centers.
- Calculating the minimum distance (Manhattan) of each parcel from public security centers.
- Calculating the minimum distance (Manhattan) of each parcel from fire stations.
- Calculating the minimum distance (Manhattan) of each parcel from emergency water supplies.
- Creating the land slope map from the existing elevation points map and extracting the slope information for each parcel from it.
- Determining the neighborhood population density number for each parcel.

TABLE 2. Equivalence of acceptable land uses for EECs in the cities of San Francisco and Tehran.

Acceptable land uses in San Francisco	Acceptable land uses in Tehran
CIE	Educational - religious
MIXED	Parking – Sports - Meeting Hall
Open Space	Green Space - Garden - Recreational (Outdoor)
Vacant	Vacant

- Removing parcels with inappropriate uses to be selected as EEC.

In the following, the required considerations are taken into account:

1) REMOVING UNACCEPTABLE LAND USES

Only certain land uses are eligible for selecting Emergency Evacuation Centers (EECs). In addition to vacant lands and green spaces that lack structural elements, land uses such as schools, mosques, stadiums, cinemas, and community halls are considered suitable. The rationale for selecting these specific land uses is tied to seismic standards for the design and construction of structures. This selection complies with the code of practice for seismic-resistant building design in Iran [53]. In this code, residential, administrative, commercial, and industrial buildings are classified as having medium importance, meaning they are designed to sustain minimal casualties during a design-level earthquake. Consequently, these buildings are not suitable to serve as Emergency Evacuation Centers (EECs) as they do not provide sufficient safety. Similarly, diplomatic properties—due to foreign government ownership—historical sites—due to their designation as world or national heritage—medical and public safety centers—due to the necessity of maintaining uninterrupted services during emergencies, as well as security concerns—are also unsuitable for accommodating the population. As a result, all these land uses have been excluded from the list of acceptable options.

Previously, when examining the data of the city of San Francisco, it was observed that the EECs have one of the four land uses CIE (Cultural, Institutional, and Educational), MIXED (Mixed Uses (Without Residential)), Open Space and Vacant. Table 2 shows the comparison of acceptable land uses in Tehran and San Francisco.

2) USING PARCELS OWNED BY THE PRIVATE SECTOR

In this research, the land ownership criterion for site-selecting EECs is not considered and all the available capacity in the city is used so that suitable parcels that are owned by the private sector are also considered as available options. In addition to the sense of philanthropy and helping the people affected by the disaster in emergencies by the private land owners, this assumption has legal support in the disaster management law of Iran, where it is announced the right to use the capacities of the private sector in the time of disaster [54].

D. DEVELOPING A NAVIGATION MODEL

To implement the population emergency evacuation process, a user-friendly model for navigating from the user's current location to the nearest EEC is implemented using web-based tools and digital map data. This model uses three key tools:

- **Leaflet:** The Leaflet JavaScript library is used to display interactive maps on the website. This library allows adding map layers from OpenStreetMap (OSM) as well as managing the display of points and routes.
- **OpenRouteService (ORS):** This is a web-based platform that provides routing services and geographic analytics using OSM data. The service includes several APIs that allow users to calculate optimal routes in different ways, including by car, bicycle, and walking. In this study, the routing option in ORS walking mode is used. By accurately calculating distance and travel time and providing route detail information, ORS enables deeper analysis and helps to optimize movement in urban spaces.
- **HTML5 Geolocation API:** This API is used to get the user's current geographic location. Using this API, the exact location of the user is identified and marked on the map.

E. ANALYSIS OF THE CASE STUDY

The application here utilizes the navigating model as discussed in Section III-D. It is evident that as the proposed EECs are based on the outputs of the AI-model, the navigating process can hence be connected and updated to the locations of the EECs. It enables people to navigate from high-risk locations to an EEC. The pseudo-code of the navigating model to increase the transparency of the research process is presented here:

1. Set up HTML structure:
 - Include Leaflet CSS and map container.
2. Initialize Leaflet map:
 - Set center and zoom.
 - Add OpenStreetMap tiles.
3. Geolocation:
 - Check for geolocation support.
 - If supported, get user location.
 - Add a marker for the user location and center the map on it.
4. Load points from a CSV file:
 - Fetch and parse CSV file.
 - Store coordinates in an array.
5. Add markers for points:
 - For each point in the array:
 - Add a marker to the map with the coordinates.
 - Attach a popup with point details.
6. Find closest point:
 - Find and store the closest point to the user's location.
7. Add routing:

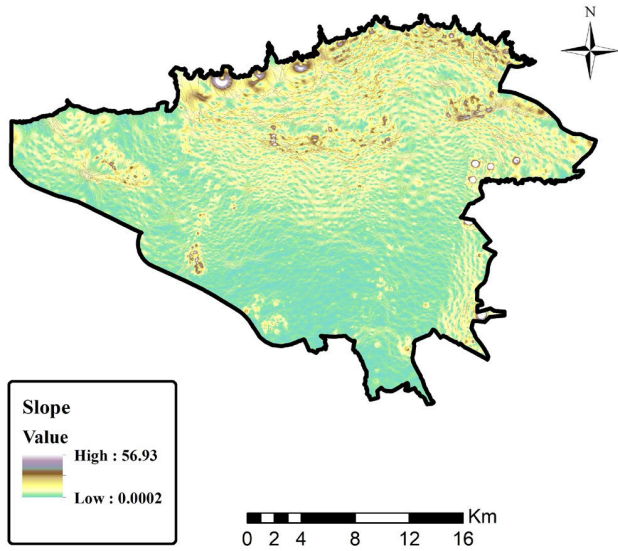


FIGURE 7. Land slope in Tehran. The map image was created by the authors using ArcGIS software version 10.7.1 (<http://www.esri.com>).

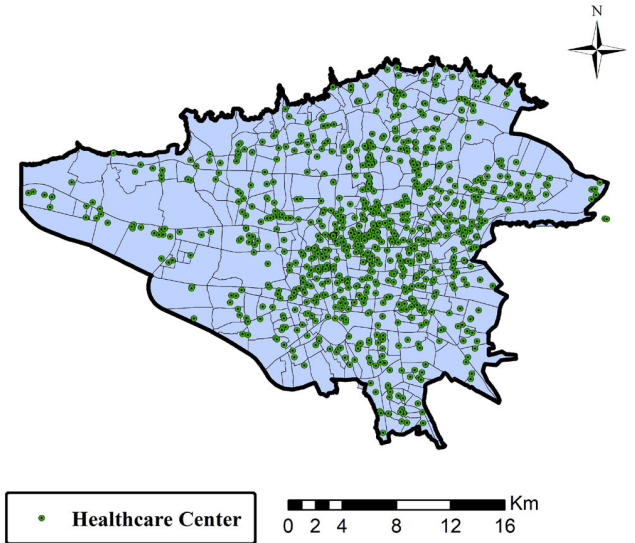


FIGURE 8. Location of medical centers in Tehran. The map image was created by the authors using ArcGIS software version 10.7.1 (<http://www.esri.com>).

- Use ORS to calculate a route from the user location to the closest point.
- Add route control to the map.

IV. RESULTS AND DISCUSSION

Here, first, in Section IV-A, the results related to the cleaning and pre-processing of the data and their details and output maps are presented. Then, in Section IV-B, the specifications of the site-selecting model, the results, and their output maps are given. In Section IV-C, the model output for the city is compared with the current condition of both Tehran and San Francisco. In Section IV-D, the previous results in Section IV-C are now developed as a navigation application.

A. INFORMATION LAYERS PREPARED FOR THE CITY OF TEHRAN

The collected information layers, due to issues such as the absence of certain data points, inconsistent coordinates, and occasionally inadequate accuracy, were cleaned and, in some cases, merged to achieve the desired quality level. It is worth noting that the amount of data after cleaning reached about 900 thousand lines (900 thousand parcels and their corresponding information); to perform the required calculations, a supercomputer with intel Xeon Gold 3.00GHz (64 Threads) CPU and 256 GB DDR4 RAM was employed. After cleaning the information, the pre-processing was conducted on the data; the outputs are shown in Figures 7-12.

B. RESULTS OF THE AI SITE-SELECTING MODEL

The information gathered from the city is converted into text format and an Excel file before being input into the Artificial Neural Network (ANN) model, which is based on San Francisco data. The model details are provided in

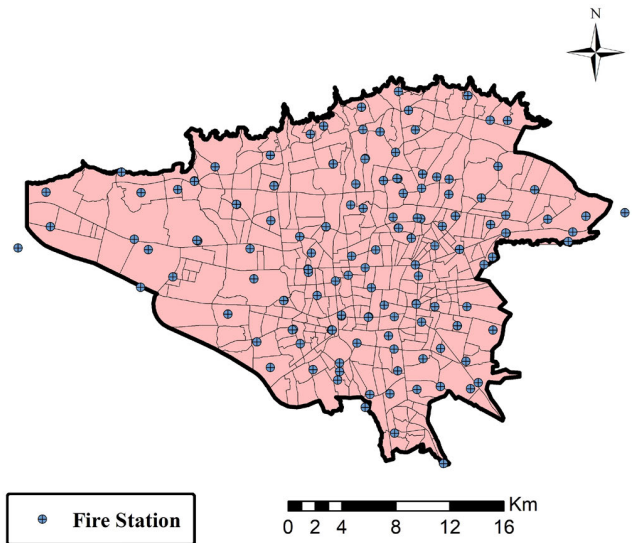


FIGURE 9. Location of fire stations in Tehran. The map image was created by the authors using ArcGIS software version 10.7.1 (<http://www.esri.com>).

Table 3. Additionally, the accuracy results of the model—trained with 80% of the San Francisco data and tested on the remaining 20%—are presented in Table 4. The F1 score is calculated using Equations 1-3, where TP and FN represent True Positive and False Negative, respectively.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

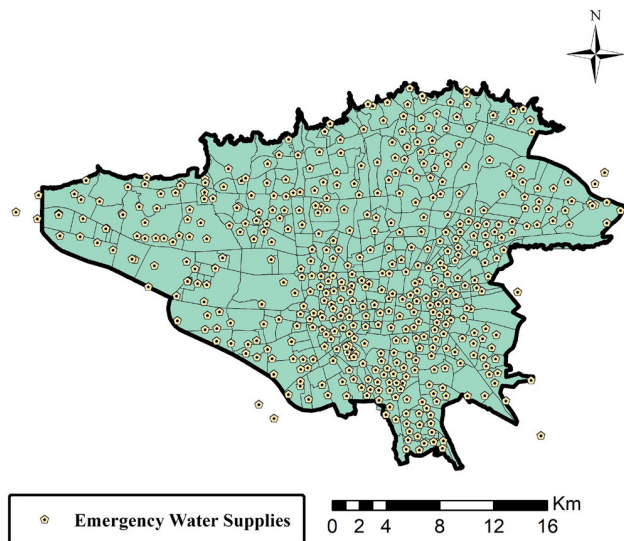


FIGURE 10. Location of emergency water supplies in Tehran. The map image was created by the authors using ArcGIS software version 10.7.1 (<http://www.esri.com>).

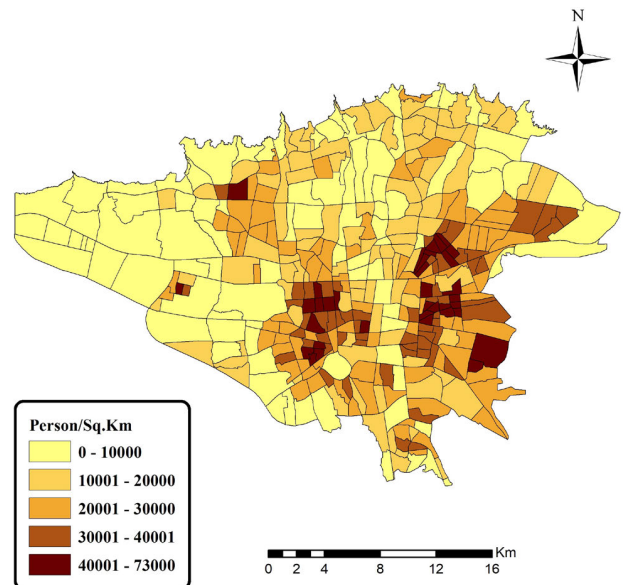


FIGURE 12. Population density of neighborhoods of Tehran. The map image was created by the authors using ArcGIS software version 10.7.1 (<http://www.esri.com>).

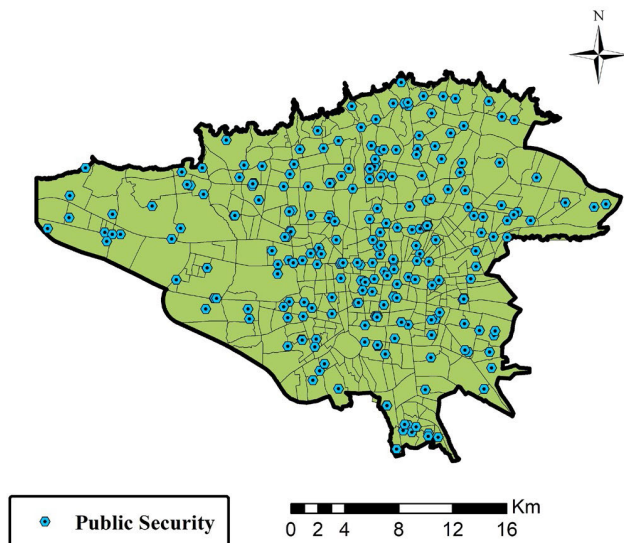


FIGURE 11. Location of public security centers in Tehran. The map image was created by the authors using ArcGIS software version 10.7.1 (<http://www.esri.com>).

TABLE 3. ANN model details.

Model	- Multi-layer Neural Network - Three hidden Dense layers with ReLU activation - An output layer with Softmax activation
Training Method	- Model trained using the fit method - Hyperparameter Tuning Phase: Trained for up to 50 epochs using EarlyStopping (patience = 5) - Final Model Training Phase: Trained for up to 100 epochs using EarlyStopping (patience = 7)
Validation Method	20% of the training data reserved as a validation set in both hyperparameter tuning (using Keras Tuner's validation_split) and final training
Hyperparameter Tuning	- Employed Keras Tuner with RandomSearch - Number of units in each hidden layer (range 8-128 with steps of 8) - Learning rate (options: 0.01, 0.001, 0.0001) - Objective: Maximize validation accuracy

TABLE 4. Confusion matrix elements and F1 score of the ANN model based on the city of San Francisco data.

TP	TN	FP	FN	F1 Score
1999	1770	163	0	0.9603

C. COMPARING THE AI-BASED RESULTS WITH THE CURRENT CONDITIONS

In this section, the results are evaluated from three perspectives. As can be seen in Table 5, the average minimum distance of each parcel to the EECs has decreased by 23%

The results including 7761 EECs are shown in Figure 13. These results are comparable in both the number and distribution of centers to those shown in Figure 3. Figures 14 and 15 show examples of suitable locations with various land uses that have been selected by the ANN model, but have not been selected as EECs in the current condition.

Except for open space land use that does not have structural elements, the logic of choosing specific land uses, as mentioned in Section III-C, is related to seismic standards for the design of urban structures. Here, the EECs must be designed to meet the requirements of the Iranian code of practice for the seismic-resistant design of buildings [53].

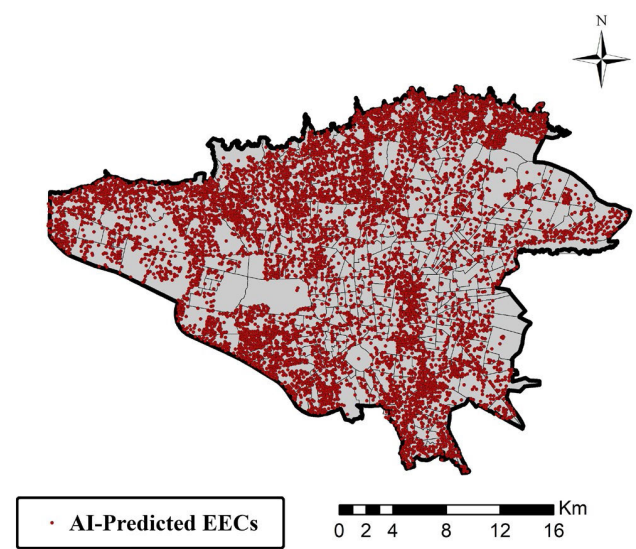


FIGURE 13. Location of EECs in Tehran as output of ANN model. The map image was created by the authors using ArcGIS software version 10.7.1 (<http://www.esri.com>).



FIGURE 14. Examples of suitable locations with diverse land uses selected by the ANN model as EECs but not currently selected as EECs in Tehran.

compared to the current conditions, and the maximum distance of the parcel to the nearest EECs has decreased by 36%, which causes faster access to EECs. The results are controlled with the design specifications for emergency shelters as newly mandated to use [55].

The next comparison is made between the site-selecting criteria and recommended values for them in some references



FIGURE 15. Examples of suitable locations with green space land use selected by the ANN model as EECs but not currently selected as EECs in Tehran.

TABLE 5. Recommended ranges for the maximum distance of a parcel to the nearest EEC and their values in the current condition and AI model output in Tehran.

	Recommended range [55]	Current Condition of Tehran	AI model output for Tehran
Maximum distance of a parcel to the nearest EEC	Class I (large areas over 20,000 m ²): 5 km	(0-2844) m Average 303.72 m	(0-1808) m Average 233.61 m
	Class II (areas between 4,000-20,000 m ²): 1 km		
	Class III (smaller areas like school sports grounds): 0.5 km		

and standards such as [4], [16], [55], [56], and [57]; the results are given in Table 6. As seen, the average values in all criteria, except the distance from emergency water supplies, are within the recommended range as given in [4], [16], [55], [56], and [57]. Regarding the criterion of distance from emergency water supplies, the average distance of EECs from emergency water supplies is close to the recommended value. However, the upper limit of the distance to emergency water supplies, which is equal to 2566 meters, is 5 times the recommended value. This shows that there is a need to increase the number of emergency water supplies, and their proper locations in the future. There are two solutions to improve accessibility to medical centers, whose response

TABLE 6. Recommended values of the criteria for site-selecting EECs in standards and their values in the current conditions and AI model output in Tehran and San Francisco.

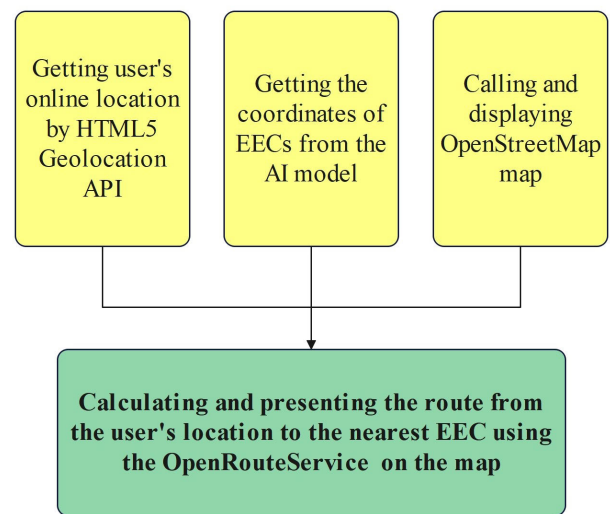
No	Criterion	Recommended value	Current Condition of Tehran	AI model output for Tehran	Current Condition of San Francisco
1	Land slope	> 8% [55] 3-4 degree [16], [56]	0-21 degree Average 2.11 degree	0-12 degree Average 2.26 degree	0.3-14 degree Average 4.37 degree
2	Distance from EEC to the fire station	4000 m [16], [56]	19-4124 m Average 1237 m	4-4287 m Average 1412 m	43-2329 m Average 771 m
3	Distance from EEC to medical center	1000 m [55] 1500 m [57]	20-2989 m Average 514 m	14-3074 m Average 738 m	50-2969 m Average 881 m
4	Distance from EEC to emergency water supply	500 m [4], [55]	1-2566 m Average 558 m	1-2566 m Average 663 m	95-1562 m Average 668 m
5	Distance from EEC to the public security center	1500 m [16], [56]	32-3830 m Average 947 m	28-3830 m Average 1113 m	157-2961 m Average 1219 m

capacity may reduce after earthquakes. As a short-term solution, mobile or filed centers can rapidly be established via which the maximum distance between already constructed medical centers reduces [58]. As a long-term solution, new properly located centers can be built. It is evident that the mentioned solutions are also applicable for public security centers.

In the third comparison perspective, the per capita EEC area in Tehran's current condition is compared to the AI-generated output; see Table 7. Given the issue of insufficient EEC capacity and the unpredictability of seismic scenarios, it is essential to prepare for the worst-case scenario, where the entire population must be accommodated. To address this, it is assumed that no fatalities occur, meaning all residents must be evacuated. The results indicate that EEC adequacy at the neighborhood level has increased from 27% in the current condition to 59% based on the AI model. Furthermore, 12 neighborhoods that previously lacked any EEC now have them. From the district perspective, the adequacy of EEC areas has improved, with the number of deficient districts decreasing from 10 to 5. The division of neighborhoods in Tehran has historical roots. Along with the city development, newer neighborhoods have emerged with different areas. Due to the official division of neighborhoods and their relative

TABLE 7. Comparison of the per capita space adequacy of EECs according to the population in the current condition of Tehran and the condition of the AI model of Tehran.

	Current Condition of Tehran	AI model output for Tehran
City neighborhoods (373 neighborhoods)	13 neighborhoods without EECs 258 neighborhoods lack EECs	1 neighborhood without EECs 150 neighborhoods lack EECs
Urban districts (22 districts)	10 districts lack EECs districts 2,7,8,9,10,11,12,13,17,18	5 districts lack EECs districts 7,8,10,11,17
According to UNHCR's announcement, the per capita EEC for the cold seasons of the year is 4.5 to 5.5 m ² for each person, and in these studies, 5 square meters per person is considered [8]		
The calculation of EEC adequacy is based on the population census of 2015, where the population of Tehran is 8.7 million [7]		

**FIGURE 16.** Navigating process from the user's location to the nearest EEC.

administrative independence, this division has been used in this research.

The comparisons show that the model developed here can make significant progress compared to the current condition of the city.

D. EMERGENCY EVACUATION NAVIGATOR

Assuming that internet access is not interrupted in emergency situations, to complete the emergency evacuation process and to make practical use of the favorable results, an application for dynamic navigating from the current location of a user to the nearest EEC is developed based on OSM map and ORS routing algorithms. The navigating process is illustrated in Figure 16. An actual example of the application's navigating is shown in Figures 17 and 18. After opening the web application, the current location is determined, and then, according to the location database of EECs, the EECs around the user's location are displayed. Next, by clicking on any of the centers,

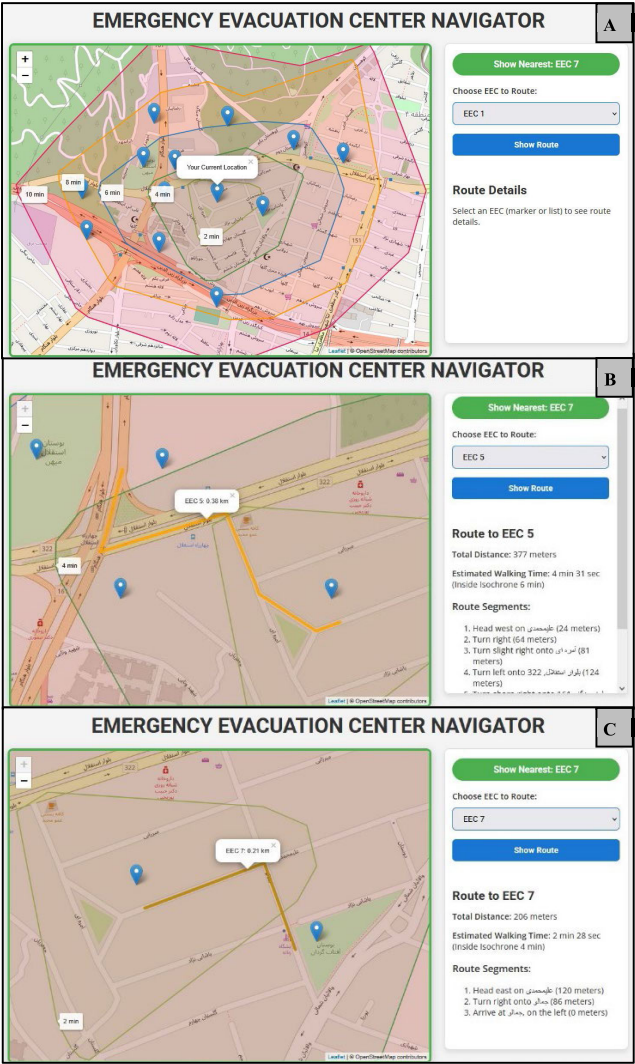


FIGURE 17. Steps of navigating from the user's location to the nearest EEC by the application. A) Showing the user's location and EECs around and isochrone zones. B) Navigating to an EEC by clicking on it or selecting from the list of nearby EECs on the right side of the page (yellow highlighted route). C) Navigating to the nearest EEC by clicking the button of show nearest EEC.

the user can get the shortest route leading to that center. It is assumed that the user can get to the center by walking and that there is no need to travel within different neighborhoods. Also, by clicking on the button of the nearest EEC, the user can immediately get directions to that center, and this route is given to the user on the map.

Here, based on a neighborhood-based approach, EECs are selected within a 1-kilometer radius of the user (approximately aligned with the boundaries of each neighborhood). This approach ensures quick and easy access for residents to the nearest EECs. Considering the average walking speed, the maximum time to reach the furthest EEC will be about 10 minutes. To show the time required to reach each center, isochrone zones are drawn to the center of the user's location, and the estimated time to reach the

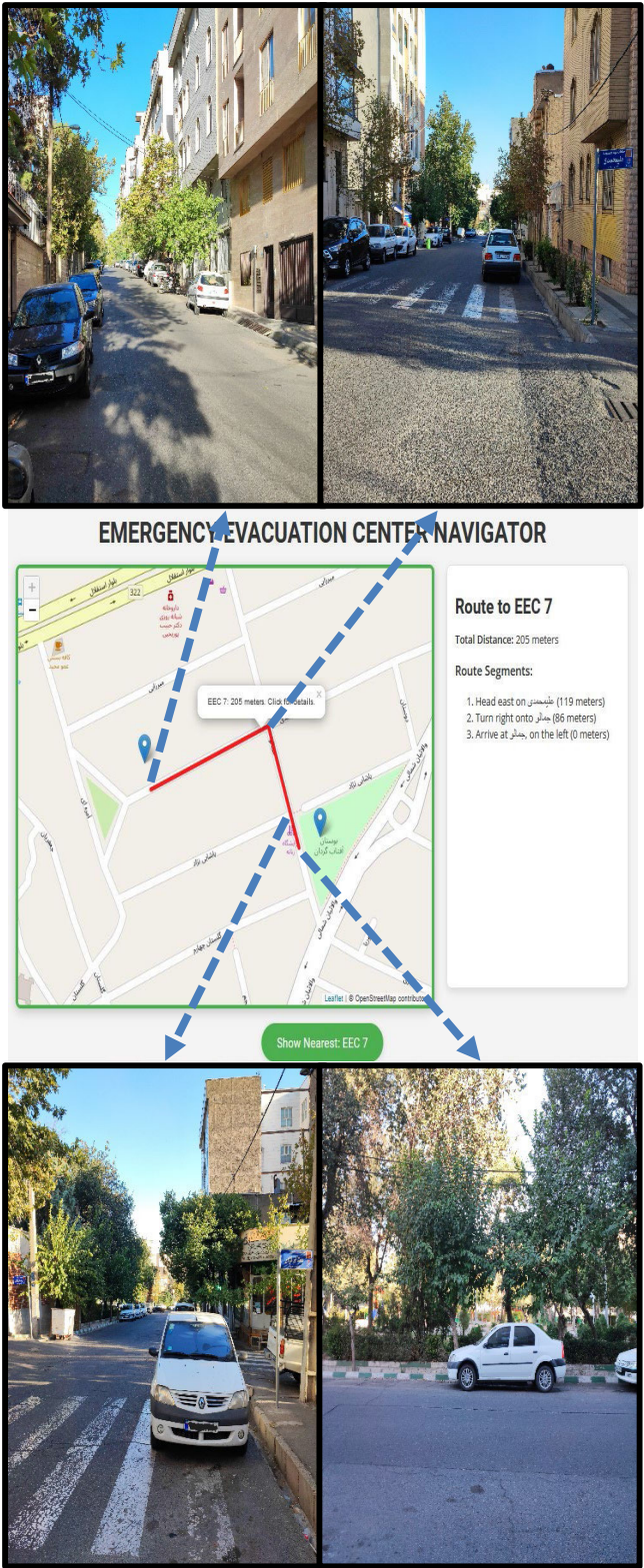


FIGURE 18. Pictures from various points along the actual sample route navigated from the user's location to the nearest EEC suggested by the application.

EEC that the user selects is provided in the route details section.

V. SUMMARY AND CONCLUSION

This research was conducted with two main objectives. The first was to develop an AI-based method to enhance the site selection of Emergency Evacuation Centers (EECs) in Tehran. The second aimed to address the existing gap in the emergency evacuation process by creating a web-based application for dynamic navigation, utilizing a neighborhood-based approach to guide users from their location to the nearest EEC.

Employing AI-based models eliminates the conventional errors associated with traditional questionnaire-based approaches. These errors include limitations in respondent sample size, spatial constraints in questionnaire results, omission of key criteria, and the static nature of questionnaire outcomes. In this research, due to the similarities between Tehran and San Francisco in terms of seismic risk, urban fabric, building types, and population density—key factors influencing the selection of Emergency Evacuation Centers (EECs)—the AI model was applied to Tehran using data from San Francisco. It is worth noting that San Francisco has extensive experience in earthquake disaster management, making it a suitable reference for developing the model for Tehran.

To achieve this, various algorithms, such as Gaussian Processes Classifier, K-Nearest Neighbors, Support Vector Machines, and Artificial Neural Networks (ANN), were tested. Ultimately, the ANN method was selected as it yielded the most favorable results, achieving an F1 score of over 95%. The ANN model was developed using information layers such as land use, land area, land slope, population density, minimum distance to emergency water supplies, minimum distance to medical centers, minimum distance to fire stations, and minimum distance to public security centers. These layers underwent necessary pre-processing before being input into the model to generate a map of the estimated EECs. The results showed:

- 1) Improving the adequacy of EECs at the neighborhood level from 27% in the current condition to 59% in the AI-based model output.
- 2) Reducing the number of neighborhoods without any EEC from 13 to just 1 in the AI model output compared to the current condition.
- 3) Reducing the number of districts lacking EECs from 10 districts to 5 districts in terms of district adequacy of EECs area.
- 4) A 23% reduction in the average minimum distance from each parcel to the EEC in the AI-based model output compared to the current condition.
- 5) A 36% reduction in the maximum distance from each parcel to the EEC in the AI-based model output compared to the current condition.

The model here can also be applied to other cities that are similar to the cities studied in terms of site-selecting criteria. For cities with less similarity, the model needs to be slightly trained with the data of the target city to achieve desirable results.

In the next step to complete the evacuation process and make practical use of the results, an application was designed and developed. This application provided dynamic navigation from the user's current location to the nearest EEC. The navigation system was based on OpenStreetMap and utilized the OpenRouteService routing algorithm.

There were limitations such as the lack of a population information layer for Tehran city during day and night, as well as the lack of updates for several layers, such as population and land use.

Also, future research can be continued in the following areas:

- Increasing the training data of the site-selecting model using data from other successful cities in the field of earthquake disaster management.
- Using more advanced neural network algorithms to increase the site-selecting accuracy.
- Adding information layers such as traffic and road damage assessment to the navigation model.
- Providing the ability to download maps and offline navigation to enable offline use of the navigation application.
- Improving the quality of walking navigation using walkability metrics (such as sidewalk quality, sidewalk width, road safety, etc.) if the required data is available.

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